

MIRC701 實驗室研究主題與計畫訪談紀錄表

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論文名稱：Distributed AI Agent-driven Framework for Coordinated Threat Detection and Response

(分散式人工智慧代理驅動框架協同威脅偵測與回應)

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一、研究動機 (Motivation):

Q1: Why did you choose distributed AI agents as your research topic? (為什麼選擇分散式 AI Agent 作為研究主題?)

Q2: What are the limitations of existing SOC architectures? (現有 SOC 架構有哪些限制?)

Q3: What is the core problem you aim to solve? (您想解決的核心問題是什麼?)

➔ Q1+Q2+Q3: SOC are operated manually by human. The centralized SOC need to handle all of the alerts sent from every security solution integrated in the system. The analyst needs to look for every alert one by one, even the redundant alerts, then they understand the progress of the attack and they can make decision on response on it. Centralized SOC is now as a bottleneck and delay the response process. That's why I want to decentralized the detection and response ability to the components in the system, and each component is operated by an AI agent that can making decision when it can act on its own, when it need the support from others

二、研究方法:

Q4: Why did you choose MITRE ATT&CK as the attack representation framework? (為何選擇 MITRE ATT&CK 作為攻擊表示方法?)

In cybersecurity field, MITRE ATT&CK provides the common knowledge of the cyberattack and it describe the process of the attack, it serve as a common language in many works in cybersecurity field

Q5: Why did you use TextCNN instead of Transformer or LLM models? (為何使用 TextCNN 而非 Transformer 或 LLM?)

I choose TextCNN because the data is not a very rich-contextual data like the language data, that's why a TextCNN can also have a good performance. Furthermore, Transformer and LLM are large models, make them more difficult to embed in the security solution

Q6: How does the leader mechanism prevent duplicate detections and conflicting responses? (Leader 機制如何避免重複偵測與衝突回應?)

Because for one attack technique, there maybe multiple agents have detection on it, and after detection the agent may give response. To avoid multiple agents become so aggressive, one agent should be a leader. The leader is the earliest agent who get the data of the attack.

Q7 : What information is exchanged among agents? (Agent 間交換哪些資訊?)

The information exchange among agents are the data they collect from the security solutions they are assigned. For example, IDS Agent collect data from IDS and do detection but IDS Agent itself is not confident enough on its detection, so it will ask other agents to provide data, the EDR Agent received the request from IDS and send its own EDR data to the IDS

三、強化學習 (Reinforcement Learning):

Q8 : Why formulate the response problem as reinforcement learning? (為什麼將回應問題建模為 RL?)

I don't formulate the response problem as RL, RL is my solution to solve the response

Q9 : How is the reward function designed? (Reward Function 如何設計?)

The reward is divide into 2 groups: reward(+) and penalty(-)

Q10 : How do you balance mitigation effectiveness and operational cost? (如何平衡防禦效果與營運成本?)

I design that the agent will get higher penalty if they choose the high-impact action in the early stages of the lifecycle, and lower the penalty of the high-impact action when the lifecycle progress deeper

四、創新與貢獻 (Innovation):

Q11 : What is the main innovation compared with previous studies? (與現有研究相比最大的創新是什麼?)

1. Decentralized the cybersecurity system to security components (in this research, the security solutions like EDR NDR, IDS, ...)
2. Put the intelligent on the security components to make them has ability to operate without the human intervention

Q12 : Why is the distributed architecture more efficient than the centralized architecture? (為何分散式架構比集中式架構更有效率?)

Distributed architecture are more efficient because when there is an cyberattack

presenting, it is no need to wait until all the data from all security solutions to be collected at centralized point, the distributed agent has the ability to react to the cyberattack within their ability and scope

Q13 : What changes would occur if LLMs were integrated in the future? (如果未來導入 LLM, 系統會有哪些改變?)

I haven' t thought about this. But I think you don' t need to present this to professor, because this is not what professor want to hear from the presentation

五、未來研究 (Future Work)

Q14 : Can this framework be applied to real enterprise environments? (此架構是否能應用於真實企業環境?)

Rightnow it is working on the lab environment, to really apply this, I think need to put more effort. And I think this is also not what professor want to hear from presentation

Q15 : Which part would you most like to improve in future work? (未來最想改進哪個部分?)

Test on more complete lifecycle, test on the environment with more agents

Q16 : Do you have any other important additions? (你有無其他重要補充?)

I suggest that for your summary presentation, you don't need to talk about the future work, it would make your presentation become longer and this part is not what professor want to listen to.